

HW 0428

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18.

(a)

percentage of fathers have some college education in this sample: 11.68 %

percentage of mothers have some college education in this sample: 12.15 %

(b)

	educ	MOTHERCOLL	FATHERCOLL
educ	1	0.359	0.398
MOTHERCOLL	0.359	1	0.355
FATHERCOLL	0.398	0.355	1

Yes, the correlations of 0.359 and 0.398 are significant, satisfying the relevance assumption and protecting the model from weak-instrument bias. MOTHERCOLL and FATHERCOLL may be superior instruments because they capture a "threshold effect" that continuous years of schooling overlook. Rather than each year of education having a constant marginal impact, a parent attaining college status often triggers a structural shift in household socioeconomic environment, access to professional networks, and academic expectations. By using dummy variables, we filter out the linear noise of schooling years and focus on the distinct, qualitative impact that a college-educated home environment has on a child's educational attainment.

(c)

95% interval estimate for the coefficient of EDUC = $[-0.0012, 0.1533]$

(d)

F-test statistic = 63.5631 > 10, MOTHERCOLL is a strong instrumental variable. This means that whether the mother has a college background strongly explains the daughter's education level, and there is no weak instrumental variable problem.

(e)

95% interval estimate for the coefficient of EDUC = [0.02752, 0.1482]

This interval is narrower than that in section (c). This is an expected result, as adding effective instrumental variables improves the accuracy of the estimate (reduces the standard error).

(f)

The F-value = 56.9629 > 10, indicating that these two variables are very robust as tool variables.

(g)

In Sargan Test, p-value = 0.626 > 0.05. We cannot reject the null hypothesis (that the instrumental variables are uncorrelated with the error term). This means that MOTHERCOLL and FATHERCOLL are valid instrumental variables, and the model specification does not exhibit significant overidentification issues.

20.

(a)

$\beta = 1.2 > 1$, This indicates that Microsoft stock is more risky than the market portfolio, exhibiting greater volatility.

(b)

RANK potentially satisfies conditions IV1—IV3. IV1 (Relevance) is supported by the empirical evidence of a high first-stage $R^2 = 0.91$ and a highly significant p -value < 2e-16. IV2 (Exogeneity) is based on the assumption that the rank is resistant to random measurement errors in the underlying values. IV3 (Exclusion Restriction) is satisfied because the rank itself has no direct economic function in explaining security returns. Given that the F-statistic of 1858 > 10, RANK can be regarded as a very strong instrumental variable.

(c)

p -value of $\hat{v} = 0.0428 > 0.01$, then the null hypothesis is NOT rejected, indicating that the market return is **exogenous**.

(d)

Yes. The IV estimate of β is 1.2783, which is notably higher than the OLS estimate. This result aligns with economic expectations because measurement error in the explanatory variable (market risk premium) typically causes attenuation bias in OLS, leading to an underestimate of the true coefficient. By using RANK as a strong instrument, the IV method corrects this downward bias, providing a consistent estimate that reflects Microsoft's status as a risky/aggressive stock relative to the market portfolio.

(e)

Yes. The joint F-statistic from the first-stage regression is 951.26, which vastly exceeds the common threshold of 10 for strong instruments. Additionally, the coefficient for RANK is highly significant ($p < 2 \times 10^{-16}$), and POS is also significant at the 5% level ($p = 0.0291$). The R^2 is 0.9149.

(f)

Yes. The coefficient of the first-stage residuals $\hat{v}_{joint}$ has a p-value of $0.0287 > 0.01$, we fail to reject the null hypothesis of exogeneity. Therefore, at the 1% level, there is insufficient evidence to conclude that the market return is endogenous, and we can treat it as exogenous.

(g)

Yes, it agrees with expectations. The increase in the coefficient magnitude suggests that the OLS estimate was biased toward zero due to measurement error in the market return (attenuation bias). By using instrumental variables, we have obtained a consistent estimate that reveals Microsoft's higher sensitivity to market fluctuations. Additionally, using two instruments instead of one has slightly improved the precision (efficiency) of our estimate compared to part (d).

(h)

The resulting p-value is $0.4549 > 0.05$, we fail to reject the null hypothesis that the instruments are valid. Therefore, we conclude that the instruments are uncorrelated with the error term and that the surplus instrument is valid for this CAPM model.